Recap Checkpoint 2 — after 1.3 (Exchanging Data) — cumulative 1.1 + 1.2 + 1.3

Name: ______________________ Date: __________ Mark: ______ / 36

OCR H446 cumulative recap — mixes every topic taught so far. Show all working.

Questions

Q1. State the function of the **MAR** and the **MDR** during the fetch phase of the FDE cycle. **[2]** *(AO1) — 1.1.1*

Q2. A laptop uses **L1, L2 and L3 cache**. Explain how cache improves CPU performance, referring to **one** difference between L1 and L3 cache. **[3]** *(AO1) — 1.1.1*

Q3. Explain how the operating system uses **interrupts and the interrupt service routine (ISR)** to respond to a key press while a program is running. **[3]** *(AO1) — 1.2.1*

Q4. A program is distributed as **bytecode** to be run by a **virtual machine**. (a) Explain **one** advantage of distributing bytecode rather than machine code. **[2]** *(AO2)*

(b) State **one** disadvantage. **[1]** *(AO2)*

[3 total] — *1.2.2*

Q5. Explain what is meant by **hashing**, and give **two** reasons why hashing is used to store passwords rather than storing the passwords directly. **[3]** *(AO1) — 1.3.1*

Q6. A row of pixels in a monochrome image is shown below, where each letter is a pixel colour:

`RRRRRGGGGGGGRRRR`

(a) Show how **run length encoding (RLE)** would encode this row. **[2]** *(AO2)*

(b) State **one** type of image for which RLE gives **little or no compression**, and explain why. **[1]** *(AO2)*

[3 total] — *1.3.1*

Q7. A company sends confidential data over the internet to a new client it has never contacted before. (a) State the difference between **symmetric** and **asymmetric** encryption. **[2]** *(AO1)*

(b) Explain why **asymmetric** encryption solves the problem of exchanging keys securely with a brand-new client. **[2]** *(AO2)*

[4 total] — *1.3.1*

Q8. A flat-file table storing orders is being normalised:

`Order(OrderID, CustomerID, CustomerName, CustomerEmail, ProductID, ProductName)`

Here `CustomerName` and `CustomerEmail` are determined by `CustomerID`, and `ProductName` is determined by `ProductID`. (a) State the rule a table must satisfy to be in **first normal form (1NF)**. **[1]** *(AO1)*

(b) Explain why this table is **not** in **third normal form (3NF)** and list the tables produced after normalising to 3NF. **[3]** *(AO2)*

[4 total] — *1.3.2*

Q9. A `Booking` table has fields `BookingID`, `Surname`, `Town`, `Nights` and `Cost`. Write an SQL statement to display the `Surname` and `Cost` of all bookings where `Town` is `'York'` **and** `Nights` is greater than `3`, sorted by `Cost` in **descending** order. **[4]** *(AO2) — 1.3.2*

Q10. Data is sent across the internet using the **TCP/IP** protocol stack. (a) Name the **four layers** of the TCP/IP stack in order from top (application) to bottom. **[2]** *(AO1)*

(b) State the layer at which the **IP address** is added, and the layer responsible for **splitting data into packets / reassembly**. **[2]** *(AO1)*

[4 total] — *1.3.3*

Q11. An online retailer's web pages load slowly and rank poorly in search results. The developers are considering moving some processing from **server-side to client-side**, and improving how pages are **indexed and ranked** by search engines. Evaluate how these two changes could help, referring to specific web technologies. **[3]** *(AO3) — synoptic 1.3.4 (+ 1.1/1.2 performance)*